Segyu Lee

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Expertise & Interest

My research focuses on building fast, cost-efficient, and scalable simulation environments to develop agents that are both capable and safe. I am particularly interested in how simulated experiences can support agent training and transfer to real-world tasks, as explored in my work on mobile GUI world models [C1] and search agent RL. My research also examines the safety of multimodal models, including their vulnerabilities and evaluation [C2, W1].

Education

Korea Advanced Institute of Science and Technology (KAIST) — Seoul, South Korea
M.S. in Graduate School of AI; Advisor: Prof. Se-Young Yun — *Sep 2025 – Ongoing*

Pohang University of Science and Technology — GPA: 3.79/4.3 — Pohang, South Korea
B.S. in Computer Science and Engineering (Major) / Entrepreneurship (Minor) — *Mar 2021 – Feb 2025*

Research Experience

Research Intern at Trillion Labs — *Dec 2025 – Jul 2026*

- Contributed to gWorld's training-data creation, benchmarks, and evaluation design; experimentally analyzed mobile GUI transition accuracy, visual fidelity, and image-generation failure modes.
- Led search-task post-training with SFT and multi-turn RL.
- Developed Simulated Scholar Search (S3), a scalable search-agent RL simulation environment based on $\sim$1M papers; studied transfer to real scholarly and web search.
- Developed a pretraining-data pipeline and generated data using paper corpora and citation graphs.

Research Intern at KAIST (OSI Lab) — *Jan 2025 – Aug 2025*

Research Intern at POSTECH (CV Lab) — *Jun 2022 – Sep 2022*

Projects

Lab Startup KAIST — *Jul 2026 – Ongoing*

Startup Team Representative

- Developing world-model-based safety guards for GUI and physical agents while leading startup and commercialization activities.

Speculative Decoding Algorithm for On-device Model — *Sep 2025 – Aug 2026*

Supported by Samsung Research

- Developed a unified evaluation framework for six speculative decoding methods and an autoregressive baseline.
- Reported 6.61$\times$ speedup (Qwen3-8B, A100, greedy decoding; eight-benchmark average).

Publications

*: 1st co-authors (equal contribution), C: conferences, W: workshops

[W1] **Segyu Lee***, B. Cho*, H. Jung*, S. An, J. Kwak, J. Kim, Y. Yang, S. Jang, Y. Park, W. Chang, and S. Yun. *"UniSAFE: A Comprehensive Benchmark for Safety Evaluation of Unified Multimodal Models"*. NeurIPS AI4GOOD Workshop (**Oral**), 2026.

[C2] J. Kwak, N. Cao, B. Cho, H. Jung, **Segyu Lee**, and S. Yun. *"When and Where to Attack? Stage-wise Attention-Guided Adversarial Attack on Large Vision Language Models"*. NeurIPS, 2026.

[C1] W. Koh*, S. Han*, **Segyu Lee**, and J. Shin. *"Generative Visual Code Mobile World Models"*. ICML, 2026.

Awards and Honors

President's Prize, Korea Insurance Development Institute (AI-based Foot Traffic Index Insurance; 2023); 3rd Place, WorldQuant Brain Alphathon (2023); Grand Prize, Asan Entrepreneurship Review Case Competition (2023); Excellence Award, Mini-I-Corps (DRT market business modeling; 2023); Excellence Award, TECHATHON (video highlight extraction; 2022); Finalist, POSTECH Startup Contest (CV-based soccer highlight extraction & Sports NFTs; 2022).